higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

BUILDING AND STRUCTURAL CONSTRUCTION N5

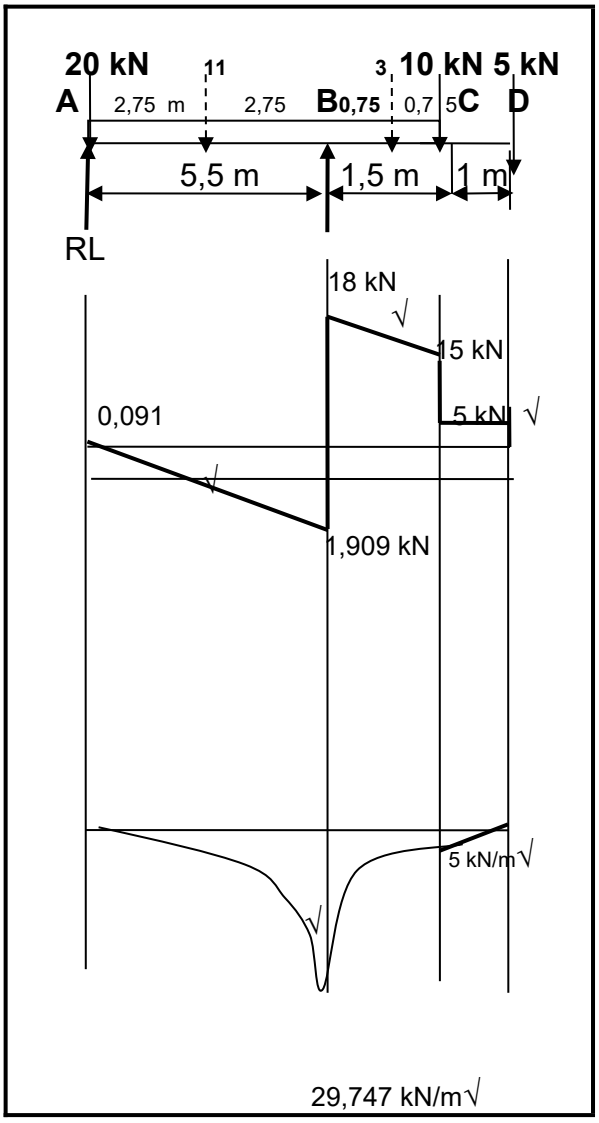
12 APRIL 2019

This marking guideline consists of 8 pages.

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SECTION A

QUESTION 1: LOADED BEAM



1.1 Take moment @RR

$RL.\frac{5,5}{5,5} = \frac{(20 \times 5,5) + (11 \times 2,75) - (3 \times 0,75) + (10 \times 1,5) - (5 \times 2,5)}{5,5}$

RL = 20,091 kN✓✓

Take moment @RL

$RR.\frac{5,5}{5,5} = \frac{(11 \times 2,75) + (3 \times 6,25) + (10 \times 7) + (5 \times 8)}{5,5}$

RR = 28,909 kN✓✓ (4)

1.2 Shear force values

@A = -20 + 20,091 = **0,091 kN**√
A-B = 0,091 – 11 = **-10,909 kN**√
@B = 10,909 – 28,909 = **18 kN**√
B-C = 18 – 3 = **15 kN**√
@C = 15 – 10 = **5 kN**√
@D = 0 (8)

1.3 Bending moment values

BM@ A = **0 kN/m**
BM@ B = (0,091 × 5,5) – (11 × 2,75) = **-29,750 kN/m**√
BM@ C = (-5 × 1) = **-5 kN/m**√
BM@ D = **0 kN/m** (4)

[16]

QUESTION 2: SELECT STEEL BEAM

2.1 Section modules

$$Ze = \frac{BM_{max}}{Pt}$$

$$= \frac{29,75\ kN/m}{155}$$

= 180 MPa✓

New Ze = 226,1✓

Choose size = 203 × 102 × 25,3 kg/m✓ (4)

2.2 Safety factors

$$F_S = \frac{Load}{Area}$$

$$= \frac{18}{203,2 \times 5,8}$$

= 15,272 MPa✓

15,272 MPa✓ less than 100 MPa is OK✓ (4)
[8]

QUESTION 3: CENTROIDS

	Area	Y-dist	A × Y	$\frac{Bh^3}{12}$ $\frac{Bh^3}{36}$ $\frac{\pi D^4}{64}$	Dist to NA	A × d^2
1	1 000	10	10 000	**33 333,33**✓	46,06	2 121 523,6
2&3	2 × 600	35	42 000	**2 × 45 000**✓	21,06	18 627 991,2
	∑2 200✓		**∑55 498**✓	**∑123 333,33**✓		**∑20 749 514,8**✓

3.1 $Y = \frac{\sum A \times Y}{\sum A}$

$Y = \frac{123\ 333,33}{2\ 200}$✓ **OR** $Y = \frac{(1\ 000 \times 10) + 2(600 \times 10)}{2\ 200}$✓

$Y = 56,06$✓ (4)

3.2 $I_{xx} = \sum (INA + Ad^2)$

$I_{xx} = \frac{(50 \times 20^3)}{12} + 2\ \frac{(20 \times 30^3)}{12}$

$\sum I_{xx} = 123\ 333,33 + 20\ 749\ 514,8\ mm^4$✓

$= 20\ 872\ 848,13\ mm^4$✓ (5)

3.3 Profile module @ NA

$Ze = \frac{I_{xx}}{Y}$

$Ze = \frac{20\ 872\ 848,13}{56,06}$✓

$Ze = 372\ 330,505 \times 10^{-6} m^3$✓ (3)

3.4 Maximum moment of resistance

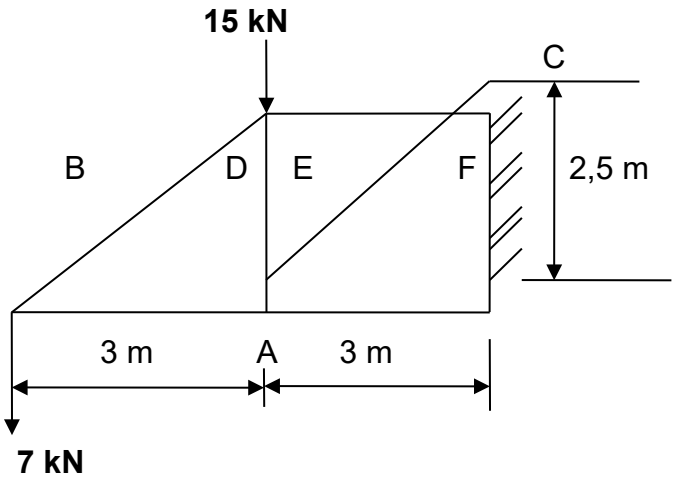
$= Ze \times Pt$

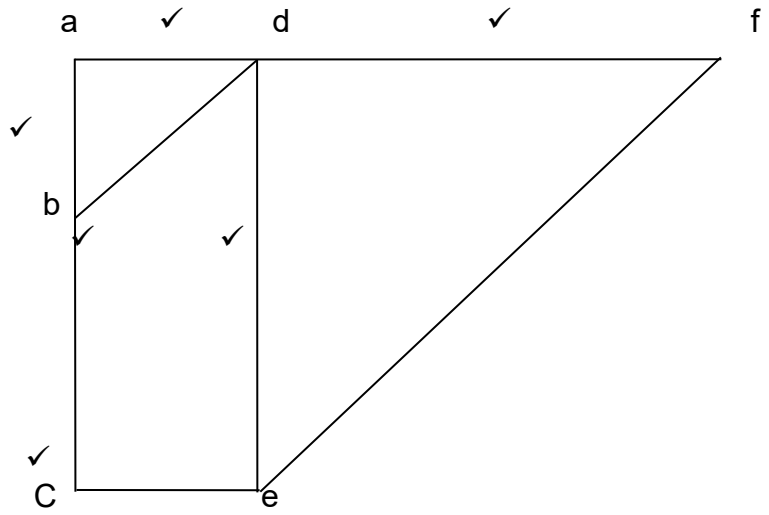
$= 372\ 330,505 \times 10^{-6} m^3 \times 165 \times 10^3$✓

$= 61,43\ kN$✓ (2)

[14]

QUESTION 4: FRAMEWORK





(6)

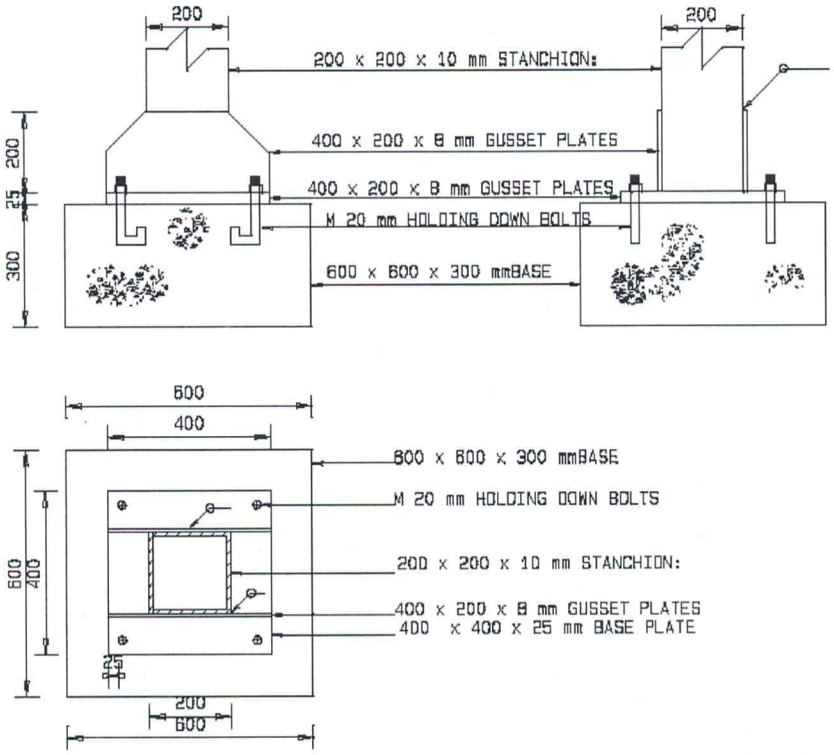
MEMBER	MAGNITUDE	NATURE
BD	Tie✓	11✓
DA	Strut✓	8✓
CE	Tie✓	8✓
ED	Strut✓	22✓
FE	Strut✓	34✓

(10)
[16]

TOTAL SECTION A: 54

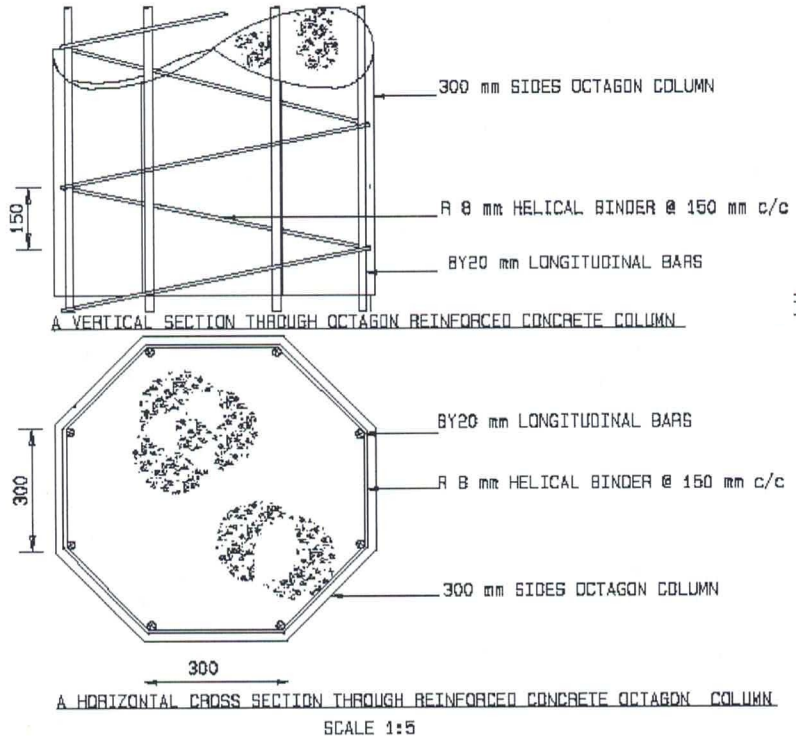
SECTION B: DRAUGHTING

QUESTION 5: STEEL PROFILES



FIRST-ANGLE PROJECTION OF A STANCHION WELDED TO A BASE PLATE
SCALE 1:5 **[16]**

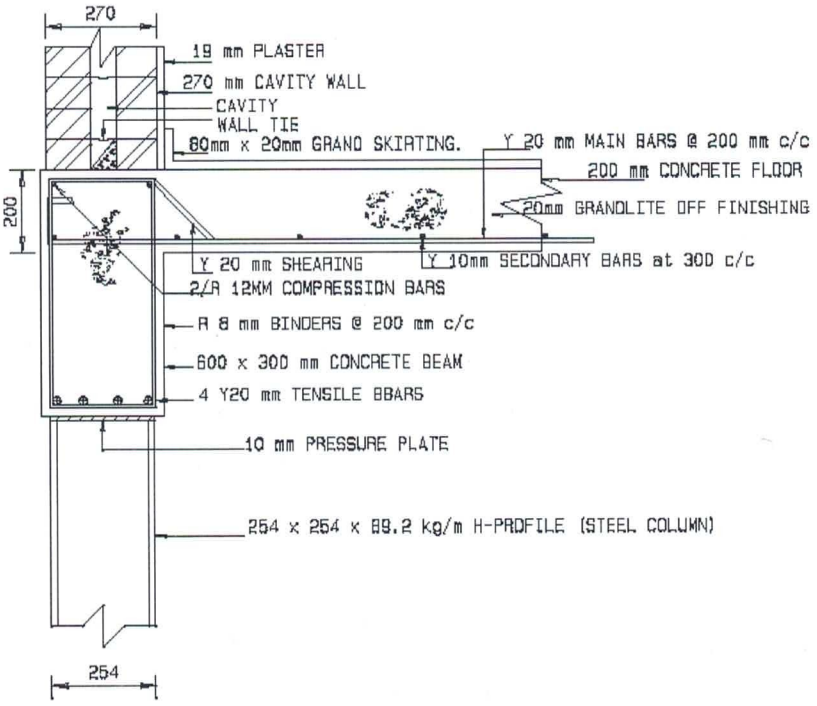
QUESTION 6: REINFORCED CONCRETE COLUMN



(5)

(5)
[10]

QUESTION 7: CONCRETE L-BEAM



STEEL COLUMN SUPPORTING A VERTICAL CROSS SECTION THROUGH A REINFORCED CONCRETE BEAM SUPPORTING A SLAB
SCALE 1:5

[20]

RUBRIC FOR QUESTIONS 5–7				
ITEM	**DESCRIPTION**	**QUESTION NUMBERS**		
		5	**6**	**7**
1	Construction accuracy	7	3	10
2	General line work	2	1	2
3	Constructional details	2	1	2
4	Labelling	3	3	4
5	Title	1	1	1
6	Scale	1	1	1
	TOTAL MARKS	16	10	20

TOTAL SECTION B: 46
GRAND TOTAL: 100

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